

Name: Solutions

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30 minutes maximum. No aids (book, calculator, etc.) are permitted. Show all work and use proper notation for full credit. Answers should be in reasonably-simplified form. 25 points possible.

1. (5 points) Consider the **telescoping** series $\sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n+2} \right)$.

(a) Find a simplified formula for S_k , the k th partial sum of the series.

$$S_k = \left(1 - \frac{1}{3}\right) + \left(\frac{1}{2} - \frac{1}{4}\right) + \left(\frac{1}{3} - \frac{1}{5}\right) + \left(\frac{1}{4} - \frac{1}{6}\right) + \cdots + \left(\frac{1}{k-2} - \frac{1}{k}\right) + \left(\frac{1}{k-1} - \frac{1}{k+1}\right) + \left(\frac{1}{k} - \frac{1}{k+2}\right)$$

$$S_k = \frac{3}{2} - \frac{1}{k+1} - \frac{1}{k+2}$$

(b) Use your answer from part (a) to determine the sum of the series.

$$\lim_{k \rightarrow \infty} \left(\frac{3}{2} - \frac{1}{k+1} - \frac{1}{k+2} \right) = \frac{3}{2} = \sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n+2} \right)$$

2. (8 points) For each **geometric** series below, determine whether the series converges and **justify** your conclusion. If the series converges, determine the sum.

$$(a) \sum_{n=1}^{\infty} \frac{3^{2n}}{4^n} = \sum_{n=1}^{\infty} \frac{(3^2)^n}{4^n} = \sum_{n=1}^{\infty} \left(\frac{9}{4} \right)^n$$

Since $r = \frac{9}{4} > 1$, the series diverges

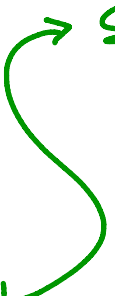
$$(b) \sum_{n=1}^{\infty} \frac{(-2)^{n-1}}{\pi 5^{n-1}} = \sum_{n=1}^{\infty} \left(\frac{1}{\pi} \right) \left(\frac{-2}{5} \right)^{n-1} = \frac{a}{1-r} = \frac{\frac{1}{\pi}}{1 - \left(\frac{-2}{5} \right)} = \frac{1}{\pi} \left(\frac{1}{\frac{3}{5}} \right) = \frac{5}{3\pi}$$

$$\text{So } a = \frac{1}{\pi}, r = \frac{-2}{5}$$

$|r| < 1$. So the series converges

3. (8 points) For each series below, determine whether the series converges and **justify** your conclusion. Your justification must include the test or series you are using to draw your conclusion.

(a) $\sum_{n=1}^{\infty} \frac{1}{n^{\sqrt{6}}}$ This is a p-series with $p = \sqrt{6} > 1$.
Thus, the series converges

(b) $\sum_{n=1}^{\infty} \frac{n}{8n+11}$
Divergence Test
 $a_n = \frac{n}{10n+17}$
 $\lim_{n \rightarrow \infty} \frac{n}{10n+17} = \frac{1}{10} \neq 0$  So, by the Divergence Test, the series diverges

4. (4 points) Apply the Integral Test to determine if the series $\sum_{n=3}^{\infty} \frac{1}{n\sqrt{\ln(n)}}$ converges or diverges. (You do not have to show that the Integral Test applies; it does.)

$$\int_3^{\infty} \frac{(\ln x)^{-1/2}}{x} dx = \lim_{t \rightarrow \infty} \int_3^t \frac{(\ln x)^{-1/2}}{x} dx = \lim_{t \rightarrow \infty} \left[2(\ln x)^{1/2} \right]_3^t$$

$$= \lim_{t \rightarrow \infty} (2\sqrt{\ln t} - 2\sqrt{\ln 3}) = \infty \quad (\text{as } t \rightarrow \infty, \ln(t) \rightarrow \infty)$$

Since the integral diverges, the series diverges.